ON THE SMALLEST SINGULAR VALUE OF WEIGHTED RANDOM MATRICES

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Abstract. Let A be an $n \times n$ random matrix with real-valued, independent, mean-zero, variance-one entries satisfying $\mathbb{E}[a_{ij}^4] \leqslant K$ for some $K > 0$, and let M be a fixed invertible $n \times n$ matrix. Writing $\tau_M = \|M^{-1}\|_{\mathrm{HS}}^{-1}$, we prove

$$\mathbb{P}\big(s_{\min}(MA) \leqslant \varepsilon \tau_M\big) \ll \varepsilon + e^{-\Omega(n)}$$

for all $\varepsilon \geqslant 0$, where the implied constants depend only on K. In the Gaussian case we also show that $s_{\min}(MA)$ is typically of size τ_M.

1. Introduction

sec:1

For an $n \times n$ matrix B, let

$$s_{\min}(B) = \min_{x \in S^{n-1}} \|Bx\|_2$$

denote its smallest singular value. Lower bounds on $s_{\min}(B)$ are used to study invertibility and the condition number. In the Gaussian case, exact formulas go back to Edelman [5,6]. For general i.i.d. ensembles, Rudelson and Vershynin first identified the $n^{-1/2}$ scale in [14] and then proved that, if the entries are subgaussian with mean zero and variance one, then

$$\mathbb{P}\big(s_{\min}(A) \leqslant \varepsilon/\sqrt{n}\big) \ll \varepsilon + e^{-\Omega(n)}.$$

This estimate is a basic result in the subject; see [15, 16]. Tao and Vu [25] obtained related bounds for random discrete matrices. Later, Rebrova and Tikhomirov [13] extended the theory to heavy-tailed i.i.d. entries, Livshyts [10] treated heavy-tailed matrices beyond the identically distributed setting, and Livshyts, Tikhomirov, and Vershynin [11] proved the same linear bound for inhomogeneous square matrices with independent entries.

In this paper we study the following random model. The matrix $A = (a_{ij})$ is an $n \times n$ random matrix with real-valued, independent, mean-zero, variance-one entries satisfying $\mathbb{E}[a_{ij}^4] \leqslant K$, and M is a fixed invertible $n \times n$ matrix. We study the product MA and ask what replaces $n^{-1/2}$ in the bound for $s_{\min}(MA)$. The answer is

$$\tau_M = \|M^{-1}\|_{\mathrm{HS}}^{-1},$$

so when M is the identity matrix, $\tau_M = n^{-1/2}$. Our main result is the following.

thm:1.1 **Theorem 1.1.** *Let $A = (a_{ij})$ be an $n \times n$ random matrix with real-valued, independent entries satisfying*

$$\mathbb{E}[a_{ij}] = 0, \qquad \mathbb{E}[a_{ij}^2] = 1, \qquad \mathbb{E}[a_{ij}^4] \leqslant K$$

for all i, j, and let M be a fixed invertible $n \times n$ matrix. Set

$$\tau_M = \|M^{-1}\|_{\mathrm{HS}}^{-1}.$$

Then

$$\mathbb{P}\big(s_{\min}(MA) \leqslant \varepsilon \tau_M\big) \ll \varepsilon + e^{-\Omega(n)}$$

for all $\varepsilon \geqslant 0$, where the implied constants depend only on K.

When M is the identity matrix, Theorem 1.1 recovers the classical bound. More generally, it shows that $\|M^{-1}\|_{\mathrm{HS}}^{-1}$ is the quantity that appears in the bound for $s_{\min}(MA)$.

Two points are worth keeping in mind. First, the theorem separates the random part from the deterministic part in a simple way. The matrix A provides the randomness, while the matrix M enters only through the single parameter τ_M. Thus $\|M^{-1}\|_{\mathrm{HS}}^{-1}$ plays the same role here that $n^{-1/2}$ plays in the usual case when M is the identity matrix.

Second, we do not assume that M has any special form beyond invertibility. Its singular values may vary widely, and M may be far from a multiple of the identity. Even in that general setting, the theorem still gives the same linear dependence on ε, with constants depending only on the fourth-moment bound for the entries of A.

There are several nearby lines of work. Earlier results on the smallest singular value for rectangular matrices and for matrices with independent columns include [1, 8]. For sparse matrices, see [3, 9]. For more global information on singular values, including fluctuation bounds in fairly general settings, see [7].

There is also a close connection with condition numbers and perturbation questions. In addition to the Gaussian work of Edelman [5, 6] and Szarek [19], one can look at the smoothed-analysis point of view in [18, 22, 24]. Related upper bounds for the smallest singular value were proved by Tatarko [26].

Another related topic is the circular law, where control of the smallest singular value is an important input. See [12, 20, 23]. For the distribution of the smallest singular value itself, see [21]. For products of deterministic and random matrices, see [27]. For a recent result on inhomogeneous rectangular matrices, see [4]. For general background on random matrices and high-dimensional probability, see [2, 28].

The fourth-moment hypothesis gives anti-concentration. Setting

$$\mathcal{L}(\xi, \varepsilon) = \sup_{u \in \mathbb{R}} \mathbb{P}(|\xi - u| \leqslant \varepsilon),$$

one has

$$\sup_{i,j} \mathcal{L}(a_{ij}, \varepsilon) \ll \varepsilon \qquad \text{for all } \varepsilon > 0,$$

where the implied constant depends only on K; see [4, Lemma 3.1].

In the Gaussian case we also prove a bound in the other direction. In particular, Theorem 5.1 shows that

$$\mathbb{P}\big(s_{\min}(MA) \leqslant C_{5.1}\, \varepsilon\, \tau_M\big) \geqslant 1 - e^{-\Omega(\varepsilon)} - e^{-\Omega(\mathrm{sr}(M^{-1}))} \qquad \text{for all } \varepsilon \geqslant 1,$$

where $\mathrm{sr}(M^{-1})$ is the stable rank of M^{-1}. For comparison, Vershynin [27] proved that, under a $(4+\varepsilon)$-th moment assumption,

$$\mathbb{E}[s_{\max}(MA)] \ll_{\varepsilon} \|M\|_{\mathrm{op}} \sqrt{n} + \|M\|_{\mathrm{HS}},$$

where $s_{\max}$ denotes the largest singular value. Thus $\|M\|_{\mathrm{HS}}$ also appears in the upper bound for $s_{\max}(MA)$.

We split vectors into compressible and incompressible ones and follow the usual distance argument. On compressible vectors we refer to Livshyts' random-rounding theorem for lower bounds on A itself. On incompressible vectors we appeal to the spread lemma from [15, Lemma 3.4], the scalar estimates from [10, Lemma 5.3, Lemma 2.2, and Proposition 3.4], and the randomized LCD bounds from [11]. The main new step is a lower bound on $\|M^{-\top} z_j\|_2$, where z_j is a measurable choice of unit vector orthogonal to the span of all columns of A except the j-th one.

subsec:1.1

1.1. **Notation and conventions.** We use the asymptotic notations $O(\cdot)$, $\Omega(\cdot)$, and $\asymp$ in the standard manner. Thus $X = O(Y)$ means that $|X| \leqslant CY$ for some constant $C > 0$, $X = \Omega(Y)$ means that $Y = O(X)$, and $X \asymp Y$ means that $X = O(Y)$ and $Y = O(X)$. In probability bounds we also write $X \ll Y$ to mean $X = O(Y)$. Unless explicitly stated otherwise, all implied constants depend only on the constant K in Theorem 1.1. When we name a constant explicitly in a lemma, proposition, or theorem, we index it by the number of that statement, for instance $c_{2.1}$ or $C_{5.1}$. Unless a statement fixes them, such constants may change from line to line. Otherwise we use the asymptotic notations above. We write $\|\cdot\|_{\mathrm{op}}$ and $\|\cdot\|_{\mathrm{HS}}$ for the operator and Hilbert–Schmidt norms, respectively. We also write $M^{-\top} = (M^{-1})^\top$, I_n for the $n \times n$ identity matrix, and $[n] = \{1, \dots, n\}$. For a nonzero matrix B, we write $\mathrm{sr}(B) = \|B\|_{\mathrm{HS}}^2 / \|B\|_{\mathrm{op}}^2$ for its stable rank. For a vector $x \in \mathbb{R}^n$, we write $\mathrm{supp}(x) = \{i \in [n] : x_i \neq 0\}$. For a vector x and a subspace H, we write $\mathrm{dist}(x, H) = \inf_{h \in H} \|x - h\|_2$. Finally, for symmetric matrices B and C, we write $B \preceq C$ when $C - B$ is positive semidefinite.

subsec:1.2

1.2. **Overview of the proof.** We divide the unit sphere S^{n-1} into compressible and incompressible vectors.

On the *compressible* side the argument is immediate: since $\|Mx\|_2 \geqslant \|M^{-1}\|_{\mathrm{op}}^{-1} \|x\|_2 \geqslant \tau_M \|x\|_2$ for every $x \in \mathbb{R}^n$, one has

$$\|MAu\|_2 \geqslant \tau_M \|Au\|_2.$$

So the weighted problem reduces to the compressible lower bound for A itself. Here we appeal to the random-rounding method of Livshyts [10].

On the *incompressible* side, write $Y_j = Ae_j$ for the j-th column of A. Suppose $u \in \mathrm{Incomp}(\delta, \rho)$ for some fixed $\delta, \rho \in (0, 1)$, and suppose $\|MAu\|_2$ is small. Then u has many spread coordinates, and for each such index j one has

$$u_j MY_j = MAu - \sum_{k \neq j} u_k MY_k,$$

so MY_j lies close to $MH_j = M\,\mathrm{span}\{Y_k : k \neq j\}$. We therefore study the formula

$$\mathrm{dist}(MY_j, MH_j) = \frac{|\langle z_j, Y_j \rangle|}{\|M^{-\top} z_j\|_2},$$

where z_j is a measurable choice of unit vector spanning $H_j^\perp$, where $H_j = \mathrm{span}\{Y_k : k \neq j\}$. For the numerator we refer to the scalar small-ball estimates in [11]. The main new step is the denominator bound

$$\mathbb{E}\|M^{-\top} z_j\|_2 = O\left(\frac{1}{\sqrt{n}\,\tau_M}\right),$$

proved by a recursive rank-reduction argument under the finite fourth-moment hypothesis.

2. The compressible bound

sec:2

For $\delta, \rho \in (0, 1)$, write $\mathrm{supp}(x) = \{i \in [n] : x_i \neq 0\}$ and

$$\mathrm{Sparse}(\delta) = \{x \in S^{n-1} : |\mathrm{supp}(x)| \leqslant \delta n\}$$

for the set of δn-sparse unit vectors, and define

$$\mathrm{Comp}(\delta, \rho) = \{x \in S^{n-1} : \mathrm{dist}(x, \mathrm{Sparse}(\delta)) \leqslant \rho\}, \qquad \mathrm{Incomp}(\delta, \rho) = S^{n-1} \setminus \mathrm{Comp}(\delta, \rho).$$

The compressible bound is an immediate consequence of a result of Livshyts [10]. Here we refer to Livshyts' random-rounding technique, which delivers the infimum over all of $\mathrm{Comp}(\delta, \rho)$ directly, without passing through a sparse bound or an operator norm estimate. This is the only input from the present section used later in the proof.

lem:2.1 **Lemma 2.1.** *Under the hypotheses of Theorem 1.1, there exist $\delta, \rho \in (0,1)$ and $c_{2.1} > 0$, depending only on K, such that*

$$\mathbb{P}\Big(\inf_{u \in \mathrm{Comp}(\delta,\rho)} \|Au\|_2 \leqslant c_{2.1}\sqrt{n}\Big) \leqslant 2e^{-c_{2.1}n}.$$

For the rest of the paper, we fix δ, ρ from Lemma 2.1 and write

$$\mathcal{C} = \mathrm{Comp}(\delta,\rho), \qquad \mathcal{I} = \mathrm{Incomp}(\delta,\rho).$$

The next proposition gives the weighted compressible bound.

prop:2.2 **Proposition 2.2.** *Under the hypotheses of Theorem 1.1, with the constants from Lemma 2.1,*

$$\mathbb{P}\Big(\inf_{u \in \mathcal{C}} \|MAu\|_2 \leqslant c_{2.1}\,\tau_M\sqrt{n}\Big) \leqslant 2e^{-c_{2.1}n}.$$

Proof. Since $\|M^{-1}\|_{\mathrm{op}} \leqslant \|M^{-1}\|_{\mathrm{HS}}$, we have

$$\|Mx\|_2 \geqslant \frac{1}{\|M^{-1}\|_{\mathrm{op}}}\|x\|_2 \geqslant \frac{1}{\|M^{-1}\|_{\mathrm{HS}}}\|x\|_2 = \tau_M\|x\|_2$$

for every $x \in \mathbb{R}^n$. In particular,

$$\|MAu\|_2 \geqslant \tau_M\|Au\|_2 \qquad \text{for all } u \in \mathbb{R}^n.$$

Taking the infimum over $u \in \mathcal{C}$, the claim now follows immediately from Lemma 2.1. □

3. The incompressible bound

sec:3

Write $Y_j = Ae_j$ for the j-th column of A. The columns $Y_1, \dots, Y_n$ are mutually independent, mean-zero, and isotropic. For each $j \in [n]$, define

$$W_j = [Y_1 \cdots Y_{j-1}\, Y_{j+1} \cdots Y_n], \qquad \mathcal{F}_j = \sigma(Y_k : k \neq j), \qquad H_j = \mathrm{span}\{Y_k : k \neq j\},$$

and set $\mathcal{R}_j = \{\dim(H_j^\perp) = 1\} = \{\mathrm{rank}(W_j) = n-1\}$. Let Π_j denote the orthogonal projection onto $H_j^\perp$, and fix an $\mathcal{F}_j$-measurable unit vector $z_j \in H_j^\perp$. On $\mathcal{R}_j$, this vector is unique up to sign.

The structure of the incompressible argument is as follows. If $u \in \mathcal{I}$ and $\|MAu\|_2$ is small, then u has a large number of spread indices, and for each such index j one has

$$u_j MY_j = MAu - \sum_{k \neq j} u_k MY_k,$$

so MY_j lies close to MH_j. The task is therefore to control, for each fixed j, the probability that the distance $\mathrm{dist}(MY_j, MH_j)$ is small. By Lemma 3.3, this distance splits into a numerator $|\langle z_j, Y_j\rangle|$ and a denominator $\|M^{-\top}z_j\|_2$. The numerator is controlled by scalar anti-concentration, while the new input is the denominator estimate in Proposition 3.4.

We first record the standard spread property of incompressible vectors from [15, Lemma 3.4].

lem:3.1 **Lemma 3.1.** *For every $x \in \mathrm{Incomp}(\delta,\rho)$ there exists a set $J(x) \subset [n]$ with $|J(x)| \geqslant \frac{1}{2}\rho^2\delta n$ such that*

$$\frac{\rho}{\sqrt{2n}} \leqslant |x_j| \leqslant \frac{1}{\sqrt{\delta n}}, \qquad j \in J(x).$$

The next lemma shows that the normal direction is well defined with overwhelming probability.

lem:3.2 **Lemma 3.2.** *For every $j \in [n]$,*

$$\mathbb{P}(\mathcal{R}_j^c) \leqslant e^{-\Omega(n)}$$

where the implied constant depends only on K.

Proof. Recall $\mathcal{R}_j^c = \{\mathrm{rank}(W_j) < n-1\}$, where W_j is the $n \times (n-1)$ matrix with columns Y_k, $k \neq j$. Let $\widetilde{W}_j$ denote the $(n-1) \times (n-1)$ submatrix of W_j obtained by deleting the last row. Its entries are a_{ik} for $i \in [n-1]$ and $k \in [n] \setminus \{j\}$, which are independent, mean-zero, variance-one, and satisfy $\mathbb{E}[a_{ik}^4] \leqslant K$. If $\widetilde{W}_j$ has rank $n-1$ then W_j has rank at least $n-1$, and since W_j is $n \times (n-1)$ its rank is at most $n-1$, so $\mathrm{rank}(W_j) = n-1$ and $\mathcal{R}_j$ holds. Therefore

$$\mathbb{P}(\mathcal{R}_j^c) \;\leqslant\; \mathbb{P}\big(s_{\min}(\widetilde{W}_j) = 0\big).$$

By the fourth-moment hypothesis and the anti-concentration estimate from the introduction, the entries of $\widetilde{W}_j$ are uniformly anti-concentrated: $\sup_{i,k} \mathcal{L}(a_{ik}, \varepsilon) \ll \varepsilon$ for all $\varepsilon > 0$. Moreover, $\mathbb{E}\|\widetilde{W}_j\|_{\mathrm{HS}}^2 = (n-1)^2 \leqslant Kn^2$. Applying [11, Theorem 1.1] to the $(n-1) \times (n-1)$ matrix $\widetilde{W}_j$ at $\varepsilon = 0$ gives

$$\mathbb{P}\big(s_{\min}(\widetilde{W}_j) = 0\big) \;\leqslant\; e^{-\Omega(n-1)} \;\leqslant\; e^{-\Omega(n)},$$

which completes the proof. □

The next lemma rewrites the weighted distance in terms of the random normal vector.

lem:3.3 **Lemma 3.3.** *On $\mathcal{R}_j$,*

$$\mathrm{dist}(MY_j, MH_j) = \frac{|\langle z_j, Y_j\rangle|}{\|M^{-\top} z_j\|_2}.$$

Proof. On $\mathcal{R}_j$, z_j spans $H_j^{\perp}$, so $\langle z_j, h\rangle = 0$ for all $h \in H_j$. For any $y = Mh \in MH_j$,

$$\langle M^{-\top} z_j, y\rangle = \langle M^{-\top} z_j, Mh\rangle = \langle z_j, h\rangle = 0,$$

so $M^{-\top} z_j \perp MH_j$. Since M is invertible, $M^{-\top} z_j \neq 0$, and therefore

$$\mathrm{dist}(MY_j, MH_j) = \left|\left\langle \frac{M^{-\top} z_j}{\|M^{-\top} z_j\|_2}, MY_j \right\rangle\right| = \frac{|\langle z_j, Y_j\rangle|}{\|M^{-\top} z_j\|_2}.$$ □

The key estimate is the following denominator bound.

prop:3.4 **Proposition 3.4.** *Under the hypotheses of Theorem 1.1, there exists $C_{3.4} > 0$, depending only on K, such that for every $j \in [n]$,*

$$\mathbb{E}[\Pi_j] \preceq \frac{C_{3.4}}{n} I_n,$$

Consequently, for every $\mathcal{F}_j$-measurable $\mathcal{A}_j \subseteq \mathcal{R}_j$,

$$\mathbb{E}\Big[\mathbf{1}_{\mathcal{A}_j} \|M^{-\top} z_j\|_2\Big] \ll \frac{1}{\sqrt{n}\,\tau_M}.$$

Proof. We prove $\mathbb{E}[\Pi_j] \preceq C_{3.4} n^{-1} I_n$ by a recursive rank-reduction argument on the columns of A.

Enumerate the columns $\{Y_k : k \neq j\}$ as $R_1, \ldots, R_{n-1}$. For $0 \leqslant m \leqslant n-1$, let P_m denote the orthogonal projection onto $\ker([R_1 \mid \cdots \mid R_m]^\top)$, so that $P_0 = I_n$ and $P_{n-1} = \Pi_j$. Set $G_m = \{\mathrm{rank}([R_1 \mid \cdots \mid R_m]) = m\}$ (with $G_0 = \Omega$), $\mathcal{G}_m = \sigma(R_1, \ldots, R_m)$, and $A_m = \mathbb{E}[P_m \mathbf{1}_{G_m}]$. On G_m, the projection P_m has rank $n-m$. Setting $\xi = R_{m+1}$, independent of $\mathcal{G}_m$, and $g = P_m \xi$, the projection updates as

$$P_{m+1} = P_m - \frac{gg^\top}{\|g\|_2^2} \mathbf{1}_{\{g \neq 0\}}, \qquad G_{m+1} = G_m \cap \{P_m \xi \neq 0\}. \tag{3.1}$$ eq:3.1

Indeed, on G_m, adjoining ξ removes from $\mathrm{Im}(P_m)$ the one direction spanned by $g = P_m \xi$, which gives the rank-one update (3.1). Since the entries of ξ satisfy $\sup_{i,j} \mathcal{L}(a_{ij}, \varepsilon) \ll \varepsilon$, the projection of

ξ onto any fixed nontrivial subspace is nonzero almost surely, so $\mathbb{P}(P_m\xi = 0 \mid \mathcal{G}_m) = 0$ on G_m. For a deterministic rank-d orthogonal projection P, write

$$B(P) = \mathbb{E}_\xi\left[\frac{P\xi\xi^\top P}{\|P\xi\|_2^2}\,\mathbf{1}_{\{P\xi\neq 0\}}\right].$$

Conditioning on $\mathcal{G}_m$ and using the almost-sure non-vanishing of $P_m\xi$ on G_m gives

$$\mathbb{E}[P_{m+1}\mathbf{1}_{G_{m+1}} \mid \mathcal{G}_m] = \mathbf{1}_{G_m}(P_m - B(P_m)),$$

and hence $A_{m+1} = \mathbb{E}[\mathbf{1}_{G_m}(P_m - B(P_m))]$.

It remains to establish a lower bound on $B(P)$. Fix $x \in S^{n-1} \cap \mathrm{Im}(P)$; since $Px = x$,

$$\langle x, B(P)x\rangle = \mathbb{E}\left[\frac{\langle x,\xi\rangle^2}{\|P\xi\|_2^2}\,\mathbf{1}_{\{P\xi\neq 0\}}\right].$$

Set $Q = \xi^\top P\xi$ and $\mathcal{A}_d = \{|Q-d| \leqslant d^{3/4}\}$. Since ξ is mean-zero and isotropic, $\mathbb{E}[Q] = \mathrm{tr}(P) = d$. Also, writing $P = (p_{ij})$ and expanding the fourth moment,

$$\mathbb{E}[Q^2] = (\mathrm{tr}\,P)^2 + 2\,\mathrm{tr}\big(P^2\big) + \sum_{i=1}^n (\mathbb{E}[\xi_i^4] - 3)p_{ii}^2.$$

Since P is an orthogonal projection, $\mathrm{tr}(P) = \mathrm{tr}\big(P^2\big) = d$ and $\sum_i p_{ii}^2 \leqslant \mathrm{tr}\big(P^2\big) = d$. Using $\mathbb{E}[\xi_i^4] \leqslant K$, we get

$$\mathbb{E}[Q^2] \leqslant d^2 + O(d),$$

and therefore $\mathrm{Var}(Q) \ll d$, so Chebyshev yields $\mathbb{P}(\mathcal{A}_d^c) \ll d^{-1/2}$. On $\mathcal{A}_d$ one has $Q \leqslant d(1+d^{-1/4})$, so dropping the $\mathcal{A}_d^c$ contribution,

$$\langle x, B(P)x\rangle \geqslant \frac{1}{d(1+d^{-1/4})}\,\mathbb{E}\big[\langle x,\xi\rangle^2\mathbf{1}_{\mathcal{A}_d}\big].$$

Since $\mathbb{E}[\langle x,\xi\rangle^2] = 1$ and $\mathbb{E}[\langle x,\xi\rangle^4] \ll 1$, Cauchy–Schwarz gives $\mathbb{E}[\langle x,\xi\rangle^2\mathbf{1}_{\mathcal{A}_d^c}] \ll d^{-1/4}$, hence $\mathbb{E}[\langle x,\xi\rangle^2\mathbf{1}_{\mathcal{A}_d}] \geqslant 1 - O(d^{-1/4})$. Hence there is a sequence $\eta_d = O(d^{-1/4})$ such that

$$\langle x, B(P)x\rangle \geqslant \frac{1-\eta_d}{d},$$

for every unit vector $x \in \mathrm{Im}(P)$. Since both $B(P)$ and P vanish on $\ker(P)$, this is equivalent to $B(P) \succeq \frac{1-\eta_d}{d}\,P$. Therefore $\sum_{d=1}^\infty \eta_d/d = O(1)$.

Since $P_m - B(P_m) \preceq \big(1 - \frac{1-\eta_{n-m}}{n-m}\big)P_m$ on G_m, we get

$$A_{m+1} \preceq \left(1 - \frac{1-\eta_{n-m}}{n-m}\right)A_m.$$

Iterating this recursion from $A_0 = I_n$ over $m = 0,\dots,n-2$, with $d = n-m$ ranging from n down to 2, gives

$$A_{n-1} \preceq \prod_{d=2}^n \left(1 - \frac{1-\eta_d}{d}\right) I_n.$$

Since $\log(1-x) \leqslant -x$,

$$\log\prod_{d=2}^n\left(1 - \frac{1-\eta_d}{d}\right) \leqslant -\sum_{d=2}^n \frac{1}{d} + \sum_{d=2}^n \frac{\eta_d}{d} = -\log n + O(1),$$

and exponentiating gives $A_{n-1} \preceq \frac{C_{3.4}}{n}\,I_n$. By Lemma 3.2, $\mathbb{P}(G_{n-1}^c) \leqslant e^{-\Omega(n)}$, and since $P_{n-1} \preceq I_n$,

$$\mathbb{E}[\Pi_j] = A_{n-1} + \mathbb{E}[P_{n-1}\mathbf{1}_{G_{n-1}^c}] \preceq \frac{C_{3.4}}{n}\,I_n + e^{-\Omega(n)}I_n \preceq \frac{C_{3.4}}{n}\,I_n,$$

absorbing the exponential error into the leading term.

For the second claim, set $Q_M = M^{-\top}M^{-1}$, so $\operatorname{tr}(Q_M) = \|M^{-1}\|_{\mathrm{HS}}^2 = \tau_M^{-2}$. On $\mathcal{R}_j$, $\Pi_j = z_j z_j^\top$, hence $\mathbf{1}_{\mathcal{A}_j} z_j z_j^\top \preceq \Pi_j$, and therefore

$$\mathbb{E}\Big[\mathbf{1}_{\mathcal{A}_j}\|M^{-\top}z_j\|_2^2\Big] = \operatorname{tr}\Big(Q_M\,\mathbb{E}[\mathbf{1}_{\mathcal{A}_j} z_j z_j^\top]\Big) \leqslant \frac{C_{3.4}}{n}\,\operatorname{tr}(Q_M) = \frac{C_{3.4}}{n\,\tau_M^2}.$$

Cauchy–Schwarz then gives

$$\mathbb{E}\Big[\mathbf{1}_{\mathcal{A}_j}\|M^{-\top}z_j\|_2\Big] \leqslant \mathbb{E}\Big[\mathbf{1}_{\mathcal{A}_j}\|M^{-\top}z_j\|_2^2\Big]^{1/2} \ll \frac{1}{\sqrt{n}\,\tau_M},$$

which proves the second claim. □

For the next step we refer to a consequence of the arithmetic part of the proof of [11, Theorem 1.2]; no further details from that argument will be needed here.

lem:3.5 **Lemma 3.5.** *There exist $\mathcal{F}_j$-measurable events $\mathcal{E}_j \subseteq \mathcal{R}_j$ with*

$$\mathbb{P}(\mathcal{E}_j^c) \leqslant e^{-\Omega(n)},$$

such that on $\mathcal{E}_j$,

$$\mathbb{P}\big(|\langle z_j, Y_j\rangle| \leqslant \varepsilon \,\big|\, \mathcal{F}_j\big) \ll \varepsilon + e^{-\Omega(n)} \qquad \text{for all } \varepsilon \geqslant 0.$$

Proof. Fix $j \in [n]$. Since $\sup_{i,k}\mathcal{L}(a_{ik},\varepsilon) \ll \varepsilon$ for all $\varepsilon > 0$, the argument in the proof of [11, Theorem 1.2] applies to the fixed column Y_j.

More precisely, we first invoke [11, Lemma 2.1] to show that, with probability at least $1-e^{-\Omega(n)}$, every unit vector orthogonal to $\{Y_k : k \neq j\}$ is incompressible. We then appeal to [11, Proposition 5.3] to obtain an $\mathcal{F}_j$-measurable event $\mathcal{E}_j \subseteq \mathcal{R}_j$ with

$$\mathbb{P}(\mathcal{E}_j^c) \leqslant e^{-\Omega(n)},$$

such that on $\mathcal{E}_j$, the unit normal z_j satisfies the RLCD lower bound required in [11, Lemma 2.5].

Condition now on $\mathcal{F}_j$. Then z_j is deterministic and Y_j is independent of $\mathcal{F}_j$. Applying [11, Lemma 2.5] with $X = Y_j$ and $v = z_j$ gives

$$\mathbb{P}\big(|\langle z_j, Y_j\rangle| \leqslant \varepsilon \,\big|\, \mathcal{F}_j\big) \ll \varepsilon + e^{-\Omega(n)}$$

for all $\varepsilon \geqslant 0$, with constants depending only on K. □

Combining the scalar small-ball bound with the denominator estimate gives the following fixed-index bound: it controls the probability that, for a given j, the column MY_j lies very close to the subspace MH_j.

cor:3.6 **Corollary 3.6.** *For every $j \in [n]$ and every $\varepsilon \geqslant 0$,*

$$\mathbb{P}\Big(\operatorname{dist}(MY_j, MH_j) \leqslant \varepsilon\sqrt{n}\,\tau_M,\ \mathcal{R}_j\Big) \ll \varepsilon + e^{-\Omega(n)}.$$

Proof. Since $\mathcal{E}_j \subseteq \mathcal{R}_j$, Lemma 3.5 gives

$$\mathbb{P}\Big(\operatorname{dist}(MY_j, MH_j) \leqslant \varepsilon\sqrt{n}\,\tau_M,\ \mathcal{R}_j\Big) \leqslant \mathbb{P}(\mathcal{E}_j^c) + \mathbb{P}\Big(\operatorname{dist}(MY_j, MH_j) \leqslant \varepsilon\sqrt{n}\,\tau_M,\ \mathcal{E}_j\Big).$$

The first term is at most $e^{-\Omega(n)}$. On $\mathcal{E}_j$, we may apply Lemma 3.3 to write

$$\operatorname{dist}(MY_j, MH_j) = \frac{|\langle z_j, Y_j\rangle|}{\|M^{-\top}z_j\|_2}.$$

Hence

$$\mathbb{P}\Big(\operatorname{dist}(MY_j, MH_j) \leqslant \varepsilon\sqrt{n}\,\tau_M,\ \mathcal{E}_j\Big)$$

$$= \mathbb{E}\Big[\mathbf{1}_{\mathcal{E}_j}\mathbb{P}\Big(|\langle z_j, Y_j\rangle| \leqslant \varepsilon\sqrt{n}\,\tau_M\,\|M^{-\top}z_j\|_2 \,\Big|\, \mathcal{F}_j\Big)\Big].$$

Applying Lemma 3.5 with $\varepsilon\sqrt{n}\,\tau_M\,\|M^{-\top}z_j\|_2$ in place of ε, and then Proposition 3.4, we obtain

$$\begin{aligned}
&\mathbb{P}\Big(\operatorname{dist}(MY_j, MH_j) \leqslant \varepsilon\sqrt{n}\,\tau_M,\ \mathcal{E}_j\Big)\\
&\quad\ll \varepsilon\sqrt{n}\,\tau_M\,\mathbb{E}\Big[\mathbf{1}_{\mathcal{E}_j}\|M^{-\top}z_j\|_2\Big] + e^{-\Omega(n)}\\
&\quad\ll \varepsilon\sqrt{n}\,\tau_M\cdot\frac{1}{\sqrt{n}\,\tau_M} + e^{-\Omega(n)} \ll \varepsilon + e^{-\Omega(n)}.
\end{aligned}$$

Combining the two bounds proves the claim. □

We can now pass from this fixed-index estimate to the full incompressible bound.

prop:3.7 **Proposition 3.7.** *For every $\varepsilon \geqslant 0$,*

$$\mathbb{P}\Big(\exists\, u \in \mathcal{I} : \|MAu\|_2 \leqslant \varepsilon\tau_M\Big) \ll \varepsilon + e^{-\Omega(n)}.$$

Proof. Define

$$\mathcal{A}_\varepsilon = \Big\{\exists\, u \in \mathcal{I} : \|MAu\|_2 \leqslant \varepsilon\tau_M\Big\}, \quad N_\varepsilon = \sum_{j=1}^{n}\Big(\mathbf{1}_{\{\operatorname{dist}(MY_j, MH_j)\leqslant(\sqrt{2}/\rho)\varepsilon\sqrt{n}\tau_M,\,\mathcal{R}_j\}} + \mathbf{1}_{\mathcal{R}_j^c}\Big).$$

Let u witness $\mathcal{A}_\varepsilon$. By Lemma 3.1, there exists $J(u) \subset [n]$ with

$$|J(u)| \geqslant \frac{1}{2}\rho^2\delta n$$

and

$$|u_j| \geqslant \frac{\rho}{\sqrt{2n}}, \qquad j \in J(u).$$

Fix $j \in J(u)$ and let $w = \sum_{k\neq j} u_k MY_k \in MH_j$. Then $u_j MY_j = MAu - w$, and since MH_j is a subspace and $0 \in MH_j$,

$$\operatorname{dist}(u_j MY_j, MH_j) = \inf_{h\in MH_j}\|MAu - w - h\|_2 = \operatorname{dist}(MAu, MH_j) \leqslant \|MAu\|_2 \leqslant \varepsilon\tau_M.$$

By homogeneity of distance from a subspace,

$$\operatorname{dist}(MY_j, MH_j) = \frac{\operatorname{dist}(u_j MY_j, MH_j)}{|u_j|} \leqslant \frac{\sqrt{2}}{\rho}\varepsilon\sqrt{n}\,\tau_M.$$

On $\mathcal{R}_j$ the first indicator contributes 1; on $\mathcal{R}_j^c$ the second indicator contributes 1. Hence

$$N_\varepsilon \geqslant |J(u)| \geqslant \frac{1}{2}\rho^2\delta n. \tag{3.2}$$ eq:3.2

Since (3.2) holds on $\mathcal{A}_\varepsilon$,

$$\mathbb{P}(\mathcal{A}_\varepsilon) \leqslant \frac{2}{\rho^2\delta n}\,\mathbb{E}[N_\varepsilon].$$

By linearity of expectation, Corollary 3.6 with $(\sqrt{2}/\rho)\varepsilon$ in place of ε, and Lemma 3.2,

$$\mathbb{E}[N_\varepsilon] \ll n\big(\varepsilon + e^{-\Omega(n)}\big).$$

Since δ and ρ depend only on K, the prefactor is $O(1)$, and therefore

$$\mathbb{P}(\mathcal{A}_\varepsilon) \ll \varepsilon + e^{-\Omega(n)}.$$ □

4. Proof of Theorem 1.1

sec:4

Proof of Theorem 1.1. For $0 \leqslant \varepsilon \leqslant c_{2.1}$, Proposition 2.2 gives, since $n \geqslant 1$,

$$\mathbb{P}\Big(\inf_{u\in\mathcal{C}} \|MAu\|_2 \leqslant \varepsilon\tau_M\Big) \leqslant 2e^{-c_{2.1}n}.$$

Decompose $S^{n-1} = \mathcal{C} \cup \mathcal{I}$. For the incompressible part, Proposition 3.7 gives

$$\mathbb{P}\Big(\exists\, u \in \mathcal{I} : \|MAu\|_2 \leqslant \varepsilon\tau_M\Big) \ll \varepsilon + e^{-\Omega(n)}.$$

A union bound over the two cases gives

$$\mathbb{P}\big(s_{\min}(MA) \leqslant \varepsilon\tau_M\big) \ll \varepsilon + e^{-\Omega(n)}$$

for $0 \leqslant \varepsilon \leqslant c_{2.1}$. For $\varepsilon > c_{2.1}$, the bound follows from $\mathbb{P}(\cdot) \leqslant 1 \leqslant c_{2.1}^{-1}\varepsilon$. □

5. Upper bound for Gaussian entries

sec:5

For the upper bound we now assume that A has i.i.d. $\mathcal{N}(0,1)$ entries. In that case a unit left singular vector corresponding to $s_{\min}(A)$ is Haar-uniform on S^{n-1} and independent of the singular values of A. This lets us combine a Hanson–Wright bound for a quadratic form with the classical Gaussian upper tail for the least singular value of a square Gaussian matrix:

$$\mathbb{P}\left(s_{\min}(A) > \frac{\varepsilon}{\sqrt{n}}\right) \;\leqslant\; e^{-\Omega(\varepsilon)}, \qquad \varepsilon \geqslant 1. \tag{5.1}$$

eq:5.1

This follows from the classical exact formulas for the smallest singular value of a square Gaussian matrix due to Edelman; see [5,6]. We prove the following theorem in this section. It shows that τ_M is the right order of magnitude for $s_{\min}(MA)$ in the Gaussian case.

thm:5.1 **Theorem 5.1.** *Let A be an $n \times n$ random matrix with i.i.d. $\mathcal{N}(0,1)$ entries, and let M be any fixed invertible $n \times n$ matrix. Write $\tau_M = \|M^{-1}\|_{\mathrm{HS}}^{-1}$ and*

$$\mathrm{sr}(M^{-1}) \;=\; \frac{\|M^{-1}\|_{\mathrm{HS}}^2}{\|M^{-1}\|_{\mathrm{op}}^2}.$$

There exists an absolute constant $C_{5.1} > 0$ such that for all $n \geqslant 1$ and every $\varepsilon \geqslant 1$,

$$\mathbb{P}\big(s_{\min}(MA) \leqslant C_{5.1}\,\varepsilon\,\tau_M\big) \;\geqslant\; 1 - e^{-\Omega(\varepsilon)} - e^{-\Omega(\mathrm{sr}(M^{-1}))}. \tag{5.2}$$

eq:5.2

In particular, when $\mathrm{sr}(M^{-1}) = \Omega(n)$, the bound (5.2) recovers the usual upper bound for $s_{\min}(A)$; compare [14,26]. The proof rests on the following quadratic-form estimate.

lem:5.2 **Lemma 5.2.** *Let N be a fixed $n\times n$ positive semi-definite matrix with $N \neq 0$, and let $v \sim \mathrm{Unif}(S^{n-1})$ be independent of N. Then*

$$\mathbb{P}\left(v^\top N v \;\leqslant\; \frac{\mathrm{tr}(N)}{4n}\right) \;\leqslant\; e^{-\Omega(\mathrm{tr}(N)/\|N\|_{\mathrm{op}})}. \tag{5.3}$$

eq:5.3

Remark. Applied with $N = M^{-1}M^{-\top}$, the identity $v^\top N v = \|M^{-\top}v\|_2^2$ and the relations $\mathrm{tr}(N) = \|M^{-1}\|_{\mathrm{HS}}^2 = \tau_M^{-2}$ and $\|N\|_{\mathrm{op}} = \|M^{-1}\|_{\mathrm{op}}^2$ give: for a Haar-uniform $v \in S^{n-1}$,

$$\mathbb{P}\left(\|M^{-\top}v\|_2 \leqslant \frac{1}{2\sqrt{n}\,\tau_M}\right) \;\leqslant\; e^{-\Omega(\mathrm{sr}(M^{-1}))}.$$

This is the denominator lower bound used in the proof. The exponent $\mathrm{tr}(N)/\|N\|_{\mathrm{op}} = \mathrm{sr}(M^{-1})$ is precisely the stable rank of M^{-1}, and it measures how spread out the eigenvalues of N are. When $\mathrm{sr}(M^{-1}) \asymp n$ (i.e., M^{-1} is nearly isotropic) the bound gives $e^{-\Omega(n)}$; when $\mathrm{sr}(M^{-1}) = 1$ (i.e., M^{-1} is rank one) the bound gives the useless constant $e^{-\Omega(1)}$, reflecting the fact that a Haar-uniform vector has only $O(1/\sqrt{n})$ projection onto a fixed one-dimensional subspace.

Proof of Lemma 5.2. Let $g = (g_1, \dots, g_n)^\top$ with $g_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, so that $v \overset{d}{=} g/\|g\|_2$ and

$$v^\top N v \;=\; \frac{g^\top N g}{\|g\|_2^2}.$$

Set $\lambda = \frac{1}{2}\operatorname{tr}(N)$. The Hanson–Wright inequality [28, Theorem 6.2.1]; see also [17]. It gives

$$\mathbb{P}\Big(g^\top N g \leqslant \operatorname{tr}(N) - \lambda\Big) \;\leqslant\; \exp\bigg(-\Omega\bigg(\min\bigg\{\frac{\lambda^2}{\|N\|_{\mathrm{HS}}^2},\; \frac{\lambda}{\|N\|_{\mathrm{op}}}\bigg\}\bigg)\bigg).$$

We claim the minimum equals $\operatorname{tr}(N)/(4\|N\|_{\mathrm{op}})$. Since $\lambda_1 \geqslant \lambda_i$ for all i (where λ_i are the eigenvalues of N), summing gives $\|N\|_{\mathrm{op}} \cdot \operatorname{tr}(N) \geqslant \|N\|_{\mathrm{HS}}^2$, hence

$$\frac{\lambda^2}{\|N\|_{\mathrm{HS}}^2} \;=\; \frac{\operatorname{tr}(N)^2}{4\|N\|_{\mathrm{HS}}^2} \;\geqslant\; \frac{\operatorname{tr}(N)}{4\|N\|_{\mathrm{op}}},$$

and trivially $\lambda/\|N\|_{\mathrm{op}} = \operatorname{tr}(N)/(2\|N\|_{\mathrm{op}}) \geqslant \operatorname{tr}(N)/(4\|N\|_{\mathrm{op}})$. Therefore

$$\mathbb{P}\Big(g^\top N g \leqslant \tfrac{1}{2}\operatorname{tr}(N)\Big) \;\leqslant\; e^{-\Omega(\operatorname{tr}(N)/\|N\|_{\mathrm{op}})}. \tag{5.4}$$ eq:5.4

Since $\|g\|_2^2 \sim \chi^2(n)$, the standard sub-exponential tail gives

$$\mathbb{P}\big(\|g\|_2^2 > 2n\big) \;\leqslant\; e^{-\Omega(n)}. \tag{5.5}$$ eq:5.5

On $\{g^\top N g \geqslant \frac{1}{2}\operatorname{tr}(N)\} \cap \{\|g\|_2^2 \leqslant 2n\}$,

$$v^\top N v \;=\; \frac{g^\top N g}{\|g\|_2^2} \;\geqslant\; \frac{\operatorname{tr}(N)}{4n}.$$

A union bound over the two failure events and (5.4)–(5.5) gives

$$\mathbb{P}\bigg(v^\top N v \leqslant \frac{\operatorname{tr}(N)}{4n}\bigg) \;\leqslant\; e^{-\Omega(\operatorname{tr}(N)/\|N\|_{\mathrm{op}})} + e^{-\Omega(n)}.$$

Since $\operatorname{tr}(N) \leqslant n\|N\|_{\mathrm{op}}$, we have $n \geqslant \operatorname{tr}(N)/\|N\|_{\mathrm{op}}$, so $e^{-\Omega(n)} \leqslant e^{-\Omega(\operatorname{tr}(N)/\|N\|_{\mathrm{op}})}$, which completes the proof. □

Proof of Theorem 5.1. Set $N = M^{-1}M^{-\top}$, so $\|M^{-\top}v\|_2^2 = v^\top N v$ for all v, $\operatorname{tr}(N) = \|M^{-1}\|_{\mathrm{HS}}^2$, and $\|N\|_{\mathrm{op}} = \|M^{-1}\|_{\mathrm{op}}^2$.

Choose a unit left singular vector $u = u_n(A)$ of A corresponding to $s_{\min}(A)$. Set $y = M^{-\top}u/\|M^{-\top}u\|_2$, which is a unit vector. Then $M^\top y = u/\|M^{-\top}u\|_2$, and since $A^\top u = s_{\min}(A)\, v_n(A)$ for the corresponding right singular vector $v_n(A)$,

$$\|(MA)^\top y\|_2 \;=\; \frac{\|A^\top u\|_2}{\|M^{-\top}u\|_2} \;=\; \frac{s_{\min}(A)}{\|M^{-\top}u\|_2}.$$

Since y is a unit vector and singular values are invariant under transpose, the variational characterization of $s_{\min}(MA) = s_{\min}((MA)^\top)$ gives

$$s_{\min}(MA) \;\leqslant\; \|(MA)^\top y\|_2 \;=\; \frac{s_{\min}(A)}{\|M^{-\top}u\|_2}. \tag{5.6}$$ eq:5.6

Since the distribution of A is invariant under left and right orthogonal multiplication, its SVD $A = U\Sigma V^\top$, with singular values arranged in nonincreasing order, has U and V Haar-distributed on $\mathrm{O}(n)$, jointly independent of Σ [2, Section 2.1.3]. In particular, $u_n(A) = Ue_n$ satisfies

$$u_n(A) \;\sim\; \mathrm{Unif}(S^{n-1}), \qquad u_n(A) \text{ independent of } s_{\min}(A). \tag{5.7}$$ eq:5.7

(Since Gaussian singular values are almost surely distinct, $u_n(A)$ is well-defined up to sign, which does not affect any subsequent quantity.)

Define

$$\mathcal{A}_\varepsilon \;=\; \left\{ s_{\min}(A) \leqslant \frac{\varepsilon}{\sqrt{n}} \right\}, \qquad \mathcal{D} \;=\; \left\{ \|M^{-\top} u_n(A)\|_2 \geqslant \frac{\|M^{-1}\|_{\mathrm{HS}}}{2\sqrt{n}} \right\}.$$

On $\mathcal{A}_\varepsilon \cap \mathcal{D}$, substituting into (5.6),

$$s_{\min}(MA) \;\leqslant\; \frac{\varepsilon/\sqrt{n}}{\|M^{-1}\|_{\mathrm{HS}}/(2\sqrt{n})} \;=\; \frac{2\varepsilon}{\|M^{-1}\|_{\mathrm{HS}}} \;=\; 2\varepsilon\,\tau_M. \tag{5.8}$$

eq:5.8

Thus Theorem 5.1 holds with $C_{5.1} = 2$, and

$$\{s_{\min}(MA) > C_{5.1}\,\varepsilon\,\tau_M\} \subseteq \mathcal{A}_\varepsilon^c \cup \mathcal{D}^c.$$

By (5.1),

$$\mathbb{P}(\mathcal{A}_\varepsilon^c) \;=\; \mathbb{P}\left(s_{\min}(A) > \frac{\varepsilon}{\sqrt{n}} \right) \;\leqslant\; e^{-\Omega(\varepsilon)}.$$

Rewrite:

$$\mathcal{D}^c \;=\; \left\{ u_n(A)^\top N\, u_n(A) \leqslant \frac{\mathrm{tr}(N)}{4n} \right\}.$$

By (5.7), $u_n(A)$ is Haar-uniform on S^{n-1}. Applying Lemma 5.2 to the deterministic matrix N,

$$\mathbb{P}(\mathcal{D}^c) \;\leqslant\; e^{-\Omega(\mathrm{tr}(N)/\|N\|_{\mathrm{op}})} \;=\; e^{-\Omega(\mathrm{sr}(M^{-1}))}. \tag{5.9}$$

eq:5.9

A union bound gives

$$\mathbb{P}\big(s_{\min}(MA) > C_{5.1}\,\varepsilon\,\tau_M\big) \;\leqslant\; \mathbb{P}(\mathcal{A}_\varepsilon^c) + \mathbb{P}(\mathcal{D}^c) \;\leqslant\; e^{-\Omega(\varepsilon)} + e^{-\Omega(\mathrm{sr}(M^{-1}))},$$

which is (5.2). □

The error term $e^{-\Omega(\mathrm{sr}(M^{-1}))}$ in Theorem 5.1 cannot, in general, be improved to $e^{-\Omega(n)}$ without a lower bound on $\mathrm{sr}(M^{-1})$. Indeed, fix any prescribed threshold $C_{5.1} > 0$. Choose $c_{5.1} > 0$, depending only on $C_{5.1}$, sufficiently small, and take

$$M^{-1} = \mathrm{diag}(n/c_{5.1}, 1, \ldots, 1).$$

Then

$$\mathrm{sr}(M^{-1}) = \frac{\|M^{-1}\|_{\mathrm{HS}}^2}{\|M^{-1}\|_{\mathrm{op}}^2} = \frac{n^2/c_{5.1}^2 + n - 1}{n^2/c_{5.1}^2} \leqslant 2, \qquad \tau_M = \frac{1}{\sqrt{n^2/c_{5.1}^2 + n - 1}} \leqslant \frac{c_{5.1}}{n}.$$

Write the rows of A as $X_1^\top, \ldots, X_n^\top$, so

$$(MA)^\top = \big[c_{5.1} X_1/n,\; X_2, \ldots, X_n\big].$$

Let $H = \mathrm{span}\{X_2, \ldots, X_n\}$. Since X_1 is independent of H, there is an orthogonal change of basis under which $(MA)^\top$ takes the form

$$\begin{pmatrix} c_{5.1} g/n & 0 \\ c_{5.1} h/n & G \end{pmatrix},$$

where $g \sim \mathcal{N}(0,1)$, $h \sim \mathcal{N}(0, I_{n-1})$, and G is an $(n-1) \times (n-1)$ Gaussian matrix, with g, h, G independent.

Consider the event

$$\mathcal{E} = \Big\{ |g| \geqslant 4(C_{5.1} + 1) \Big\} \cap \Big\{ \|h\|_2 \leqslant 2\sqrt{n} \Big\} \cap \Big\{ s_{\min}(G) \geqslant 8(C_{5.1} + 1) c_{5.1}/\sqrt{n} \Big\}.$$

Since g, h, G are independent, the first two events have probability bounded below by positive constants depending only on $C_{5.1}$. By the standard lower-tail bound for the smallest singular value of an $(n-1) \times (n-1)$ Gaussian matrix, after decreasing $c_{5.1}$ in terms of $C_{5.1}$ if necessary, the third

event also has probability bounded below by a positive constant depending only on $C_{5.1}$. Hence, after decreasing $c_{5.1}$ once more if necessary,

$$\mathbb{P}(\mathcal{E}) \geqslant c_{5.1}.$$

This lower bound is uniform in n.

Now let $x = (x_1, x') \in S^{n-1}$. On $\mathcal{E}$ we have

$$\|(MA)^\top x\|_2^2 = \left|\frac{c_{5.1} g}{n} x_1\right|^2 + \left\|\frac{c_{5.1} h}{n} x_1 + Gx'\right\|_2^2.$$

If $\|x'\|_2 \leqslant 1/2$, then $|x_1| \geqslant \sqrt{3}/2$, and hence

$$\|(MA)^\top x\|_2 \geqslant \frac{c_{5.1}|g|}{n}|x_1| \geqslant \frac{2(C_{5.1}+1)c_{5.1}}{n} \geqslant \frac{2C_{5.1}c_{5.1}}{n}.$$

If $\|x'\|_2 \geqslant 1/2$, then

$$\|(MA)^\top x\|_2 \geqslant s_{\min}(G)\|x'\|_2 - \frac{c_{5.1}\|h\|_2}{n}|x_1| \geqslant \frac{4(C_{5.1}+1)c_{5.1}}{\sqrt{n}} - \frac{2c_{5.1}}{\sqrt{n}} = \frac{(4C_{5.1}+2)c_{5.1}}{\sqrt{n}} \geqslant \frac{2C_{5.1}c_{5.1}}{n}.$$

Therefore, on $\mathcal{E}$,

$$\|(MA)^\top x\|_2 \geqslant \frac{2C_{5.1}c_{5.1}}{n} \qquad \text{for all } x \in S^{n-1}.$$

Since $\tau_M \leqslant c_{5.1}/n$, this gives

$$s_{\min}(MA) \geqslant C_{5.1}\,\tau_M \qquad \text{on } \mathcal{E}.$$

Consequently,

$$\mathbb{P}\big(s_{\min}(MA) > C_{5.1}\,\tau_M\big) \geqslant c_{5.1} > 0$$

for all n, while $\mathrm{sr}(M^{-1}) \leqslant 2$. Thus no estimate of the form

$$\mathbb{P}\big(s_{\min}(MA) \leqslant C_{5.1}\,\tau_M\big) \geqslant 1 - e^{-\Omega(n)}$$

can hold uniformly over all invertible M. In particular, the dependence of the error probability on $\mathrm{sr}(M^{-1})$ in Theorem 5.1 is genuine.

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